School of Informatics & IT
TEMASEK POLYTECHNIC

Data Warehousing and Business Intelligence

CDA2C01

Topic 3
Data Integration and Data Warehouse Implementation

Learning Outcomes

- ✓ Explain the integration of data to create a data warehouse
- ✓ Describe Extract Transform and Load (ETL) concepts for Big Data.
- ✓ Design ETL task.

MAJOR STEPS IN THE ETL PROCESS

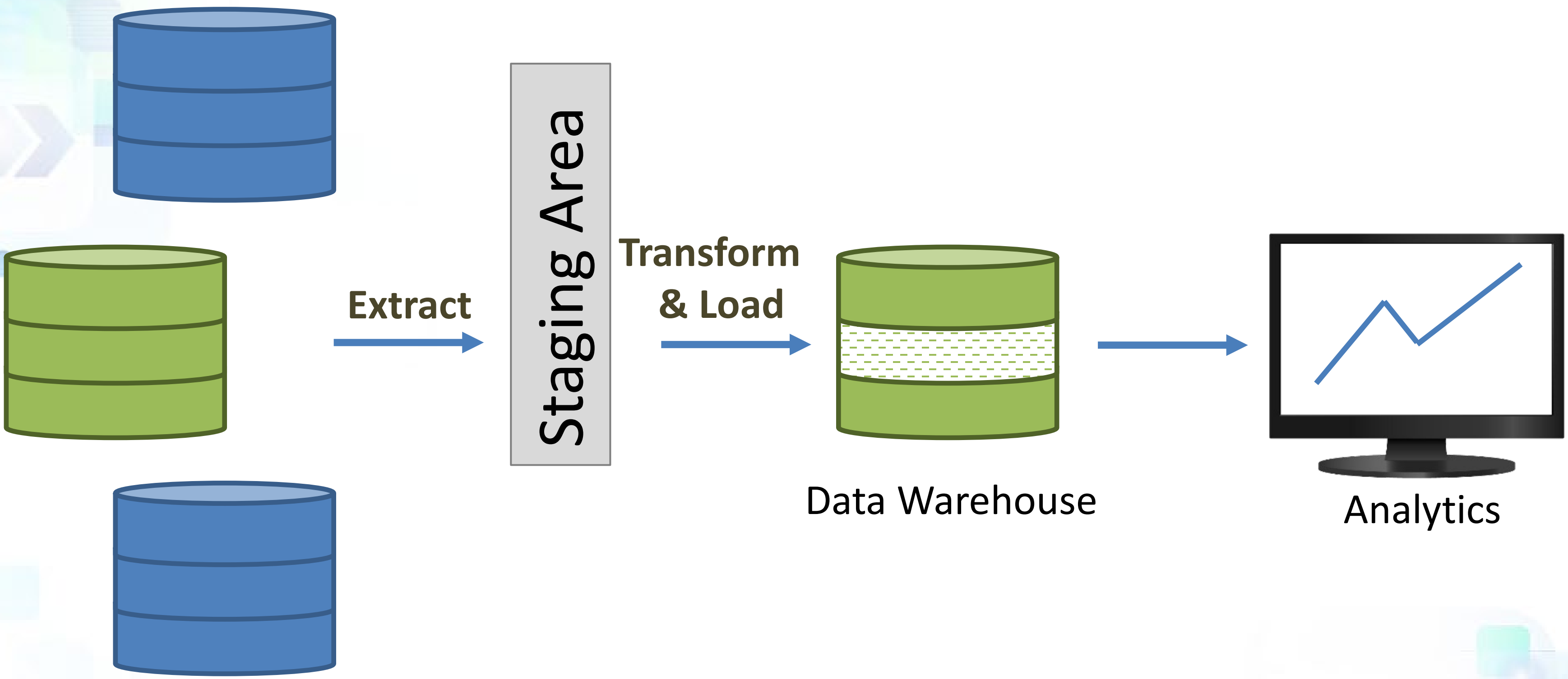
WHAT IS ETL?

ETL is a type of data integration that refers to the three steps (extract, transform, load) used to blend data from multiple sources.

It's often used to build a data warehouse.

During the process, data is taken (extracted) from a source system, converted (transformed) into a format that can be analysed,. and stored (loaded) into a data warehouse

WHAT IS ETL?



Extract
Staging Area
Transform
& Load
Data Warehouse
Analytics

BENEFITS OF ETL TOOLS

- Ease of Use
- Visual Flow
- Advanced Data Profiling and Cleansing
- Enhanced Business Intelligence
- High Return on Investment (ROI)

EXAMPLE OF ETL TOOLS

- Cognos Decision Stream
- Oracle Warehouse Builder
- Business Objects XI
- SAS business warehouse
- SAS Enterprise ETL server
- TIBCO

Major steps in the ETL process

- ETL for fact tables
- ETL for dimension tables
- Write procedures for all data loads
- Organize data staging area and test tools
- Plan for aggregate tables
- Determine data transformation and cleansing rules
- Establish comprehensive data extraction rules
- Prepare data mapping for target data elements from sources.
- Determine all the data sources, both internal and external
- Determine all the target data needed in the data warehouse

Additional on ETL

DATA EXTRACTION ISSUES

- Source identification—identify source applications and source structures.
- Method of extraction—for each data source, define whether the extraction process is manual or tool-based.
- Extraction frequency—for each data source, establish how frequently the data extraction must be done: daily, weekly, quarterly, and so on.
- Time window—for each data source, denote the time window for the extraction process.
- Job sequencing—determine whether the beginning of one job in an extraction job stream has to wait until the previous job has finished successfully.
- Exception handling—determine how to handle input records that cannot be extracted.

DATA TRANSFORMATION

Note for Project

- Basic Tasks
  - Selection
  - Splitting / Joining
  - Conversion
  - Format Revisions
  - Decoding of Fields

MAJOR TRANSFORMATION TYPES

Note for Project

- Enrichment
- Calculated and Derived Values
- Splitting of Single Fields
- Merging of Information
- Character set conversion
- Summarization
- Key Restructuring
- Deduplication

DATA LOADING

- Initial load—populating all the data warehouse tables for the very first time.
- Incremental load—applying ongoing changes as necessary in a periodic manner.
- Full refresh—completely erasing the contents of one or more tables and reloading with fresh data (initial load is a refresh of all the tables).

ELT? ETL?